

Trios Cyber Internship — Day 01 Technical Report

Kali Linux & Network Setup

Organization: Trios Cyber

Internship: Cybersecurity Internship

Task: Kali Linux & Network Setup

Day: 01

Date: 01 September 2026

Environment: Kali Linux Virtual Machine

Virtualization Platform: Oracle VirtualBox

Network Mode: NAT

Status: Completed

1. Executive Summary

The Day 01 task focused on establishing and validating a Kali Linux virtual laboratory environment and performing basic system and network enumeration.

The assessment covered system identification, network interface enumeration, IP address identification, routing analysis, connectivity verification, and listening-service enumeration.

All required commands were executed in the Kali Linux lab environment. Command outputs were preserved as raw log files, and screenshots were captured as evidence.

2. Objectives

The objectives of this task were to:

1. Verify the Kali Linux hostname.
 2. Identify the current user.
 3. Enumerate available network interfaces.
 4. Identify the lab/attacker IP address.
 5. Identify the default gateway.
 6. Inspect the routing table.
 7. Verify local network connectivity.
 8. Verify external network connectivity.
 9. Enumerate listening TCP/UDP services.
 10. Document the results using screenshots and command logs.
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3. Lab Environment

Parameter	Value
Operating System	Kali Linux
Environment	Virtual Machine

Parameter	Value
Virtualization	Oracle VirtualBox
Network Mode	NAT
Primary Interface	eth0
Lab/Attacker IPv4	10.0.2.15/24
Default Gateway	10.0.2.2
Loopback Address	127.0.0.1

4. System Identification

4.1 Hostname

Command:

```
hostname
```

Result:

```
kali
```

The hostname of the Kali Linux system was confirmed as **kali**.

Evidence: screenshots/01-hostname.png

4.2 Current User

Command:

```
whoami
```

Result:

```
yuvaraj
```

The current logged-in user was confirmed as **yuvaraj**.

Evidence: screenshots/02-whoami.png

5. Network Interface Enumeration

5.1 IP Address and Interface Configuration

Command:

```
ip a
```

The primary network interface identified was:

```
eth0
```

The assigned IPv4 address was:

10.0.2.15/24

The loopback interface was:

lo

127.0.0.1/8

The `eth0` interface was operational and configured with the 10.0.2.0/24 network.

Evidence: screenshots/03-ip-a.png

5.2 Legacy Interface Information

Command:

`ifconfig`

The `ifconfig` output confirmed:

Interface: `eth0`

IPv4 Address: 10.0.2.15

Netmask: 255.255.255.0

Broadcast: 10.0.2.255

The interface reported no RX or TX errors during the captured observation.

Evidence: screenshots/04-ifconfig.png

6. Routing Analysis

6.1 Routing Table

Command:

`ip route`

The default route was identified as:

`default via 10.0.2.2 dev eth0`

The local network route was:

`10.0.2.0/24 dev eth0`

Routing Findings

Parameter	Value
Default Gateway	10.0.2.2
Interface	<code>eth0</code>
Local Network	10.0.2.0/24
Source IP	10.0.2.15

The routing configuration indicates that traffic outside the local 10.0.2.0/24 network is forwarded through 10.0.2.2.

Evidence: screenshots/05-ip-route.png

7. Connectivity Verification

7.1 Gateway Connectivity

The default gateway was tested using ICMP:

```
ping -c 4 10.0.2.2
```

Result:

```
4 packets transmitted
```

```
4 packets received
```

```
0% packet loss
```

Average round-trip time:

```
0.445 ms
```

This confirmed successful communication between the Kali VM and its configured gateway.

Raw output: logs/ping-gateway.txt

7.2 External Network Connectivity

External connectivity was tested using Google's public DNS server:

```
ping -c 4 8.8.8.8
```

Result:

```
4 packets transmitted
```

```
4 packets received
```

```
0% packet loss
```

Average round-trip time:

```
7.590 ms
```

This confirmed that the Kali VM had functional external network connectivity.

Raw output: logs/ping-internet.txt

Evidence: screenshots/06-ping.png

8. Listening Service Enumeration

Listening TCP/UDP services were checked using the socket listing output.

The captured result contained only the header:

```
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
```

No listening services were reported during the observation.

This indicates that no TCP/UDP listening sockets were detected at the time of enumeration.

Raw output: logs/listening-services.txt

Evidence: screenshots/07-listening-services.png

9. Network Configuration Summary

Item	Finding
Hostname	kali
Current User	yuvaraj
Primary Interface	eth0
Lab/Attacker IPv4	10.0.2.15/24
Local Network	10.0.2.0/24
Default Gateway	10.0.2.2
Loopback	127.0.0.1
Gateway Connectivity	Successful
Gateway Packet Loss	0%
External Connectivity	Successful
External Packet Loss	0%
Listening Services	None reported

10. Evidence and Documentation

The following evidence was collected during the task.

Screenshots

File	Evidence
01-hostname.png	Hostname verification
02-whoami.png	Current user verification
03-ip-a.png	IP addresses and interfaces
04-ifconfig.png	Interface configuration
05-ip-route.png	Routing table
06-ping.png	Connectivity testing
07-listening-services.png	Listening-service enumeration

Raw Logs

File	Description
day1-network-enumeration.txt	Main enumeration output
network-summary.txt	Network configuration summary
listening-services.txt	Listening-service enumeration
ping-gateway.txt	Gateway connectivity test
ping-internet.txt	External connectivity test

11. Commands Practiced

The following Linux networking and system-identification commands were used:

```
hostname
whoami
ip a
ifconfig
ip route
ping -c 4 10.0.2.2
ping -c 4 8.8.8.8
```

Listening services were also enumerated using the socket listing functionality.

These commands provided practical exposure to system identification, interface discovery, routing analysis, connectivity testing, and basic service enumeration.

12. Conclusion

The Day 01 Kali Linux and Network Setup task was completed successfully in a dedicated VirtualBox-based Kali Linux laboratory environment.

The Kali VM was identified, its network interfaces and IP configuration were enumerated, the default gateway and routing table were identified, and network connectivity was successfully verified.

The environment demonstrated:

- A functional `eth0` interface.
- Lab IP address `10.0.2.15/24`.
- Default gateway `10.0.2.2`.
- Successful gateway connectivity with 0% packet loss.
- Successful external connectivity with 0% packet loss.
- No listening TCP/UDP services reported during the observation.

All required evidence was organized into screenshots and raw command-output logs for reproducibility and review.

13. Repository Structure

```
Day-01-Kali-Network-Setup/  
  README.md  
  DAY-01-REPORT.md  
  logs/  
    day1-network-enumeration.txt  
    listening-services.txt  
    network-summary.txt  
    ping-gateway.txt  
    ping-internet.txt  
  screenshots/  
    01-hostname.png  
    02-whoami.png  
    03-ip-a.png  
    04-ifconfig.png  
    05-ip-route.png  
    06-ping.png  
    07-listening-services.png
```

Prepared for: Trios Cyber Internship

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